IoTBay Marketplace
Project Brief

A. Project Description & Objectives

As an Internet of Things (IoTs) business, IoTBay has a wide range of collections of sensors, actuators, gateways, etc. IoTBay currently uses a desktop application for managing customer details, inventory, orders and payments in store, but is unable to keep up with the growing amount of data and rapid increase in the demand for specialized devices from customers who are unable to visit the store in person. IoTBay aims to improve their business portfolio by providing a user-friendly, online marketplace that achieves the following objectives:

1. Improve inventory management of IoT devices
2. Enabling online browsing and purchasing
3. Improve user information management

B. Project Scope

The key users will be IoTBay staff and customers. The website will allow customers to register, login/logout, manage their own accounts, payment methods, payments, and orders. The website will allow staff to manage the inventory, customer orders and shipments. The system also stores user access logs. More detailed descriptions of each feature are provided in the IoTBay Key Features table. **This table does not cover all details of the system's operation, leaving rooms for reasonable extensions and design choices.**

Table: IoTBay Key Features

ID	Feature	Create	Read	Update	Delete
00	User Registration & Authentication	A user (a customer or staff) can register through a registration form. During registration, the email and password formats should be validated. A user has 2 statuses: active or inactive. Upon registration, the user's status is set to active by default. During login, credentials should be validated against database records. The required fields are different for staff and customers: staff records may include name, email, staff id, position, etc, while user records may include full name, email, phone number, and address (if feature Shipment Management is selected) etc.	A registered user can view their registration details. A registered user can log into and log out from the system.	A registered user can update their registration details.	A registered user can cancel their registration. When a user account is cancelled, the user remains in the database, the user's status is changed to inactive, any stored payment methods are deleted, all unpaid orders are marked cancelled, shipments are not affected.
01	User Access Management	A registered user access logs (user id, login timestamps, logout timestamps, datetime,	A registered user can view their access logs and search the log records by date.	A registered user cannot update their user access logs.	A registered user cannot delete their user access logs.

	This applies to both customer and staff user.	etc) are stored in the database.			
02	IoT Device Catalogue Management	Only staff user can create IoT device details (e.g., device name, type, unit price, quantity/stock) in the database.	Customer and staff users can view the IoT device catalogue. Customer and staff users can search and view the devices based on their name and type.	Only staff user can update the IoT device records.	Only staff user can delete IoT device records.
03	Orders	A registered customer user can create an order for IoT devices. Statuses of orders include saved, paid, cancelled. When an order is placed, its status is set to saved, and the number of devices should decrease accordingly. An order can contain several types of devices and includes the total price. Customers should not be able to order a device that has 0 stock.	A customer user can view all their orders and search by order id and date. A staff user can view all customer orders.	A customer user can update their saved orders before they are paid.	A customer can cancel their saved order before they are paid. When an order is cancelled, its status is updated to 'cancelled', but the order remains in the database; the number of devices should be added back accordingly.
04	Payment Methods	A customer can create payment methods (e.g., credit card, bank account transfer, etc). Each payment type should contain fields relevant to that specific type. Payment methods should have unique identifiers (payment method id). Design a simple algorithm to check if the entered payment method is valid.	A customer user can view their own payment methods and search by type, etc. A staff user cannot view a customer's payment methods.	A customer user can update their payment methods. A staff user cannot update a customer's payment methods.	A customer user can delete their payment methods. A staff user cannot delete a customer's payment methods.
05	Payments	To pay for an order, a customer creates a payment by selecting an existing order and an existing payment method. Both the order and the payment method should already exist in the database. The order should have a status "saved". This changes the status of the order to paid. The payment item should also contain the amount and the date of payment.	A customer user can view their own payment history; a staff user can view all customer's payments.	Neither a customer user nor a staff user can update the payment history.	Neither a customer user nor a staff user can delete the payment history.
06	Shipments	Staff can create shipment linked to an order id once the order is paid. Only	Shipment can be viewed by the customer who places	The "assigned" staff user can update the	A shipment cannot be deleted by a

		one shipment can exist per order. The shipment can have statuses pending, shipped, delivered, which is updated by staff. The shipment retrieves address from the customer's record. Once the shipment is created by a staff member, that staff member is considered "assigned" to the shipment.	the order and the "assigned" staff user.	shipment details and status. The customer cannot update the shipment.	customer or staff user.
07	Shopping Cart	A customer always has a shopping cart. The shopping cart reserves items selected.	A customer can view all items in their own shopping cart.	A customer can add or remove items from the shopping cart. The shopping cart can be emptied by removing all items manually, or automatically when the customer clicks the "create an order" button, which also generates an order containing all items in the cart.	A customer cannot delete the shopping cart.
08	Built-in Admin User Workflow Triage	The admin can create new user accounts (customer or staff), devices, orders, payments, shipments. The admin cannot create user access logs or payment methods. Paid orders are visible to the admin and are, by default, hidden from staff members. The admin can assign paid orders to staff members for creating shipments, which grants the specific staff member access to the order.	The admin can view all user accounts, devices, access logs, orders, payments, shipments. The admin cannot view customers' payment methods.	The admin can update details of users, devices, orders, shipments. The admin cannot edit user access logs or customers' payment methods, payments.	The admin cannot delete user access logs, orders, shopping cart, payment methods, payments, shipments.

*** Customer id, staff id, order id, shipment id, etc., should be unique and may be autogenerated.**

C. Project Team

You will form groups of **4-6 students with others in the same workshop session by the end of Week 4**. Each group should nominate a **project lead**.

The client wants the software for review in two releases, with **R0 implementing Feature 00** and **R1 implementing mandatory Features 01, 02, 03, and 04**. Groups with more than four members will select **additional features from Features 05-08** to ensure that each member individually implements at least one feature. Groups with fewer than four members may be permitted to work on a reduced scope, in consultation with the workshop tutor.